Subjective expectations, Stated preferences, and Co.

Romuald Méango

Motivating example I

Think of the question of career choice: for example, the decision of an individual to work for the public or the private sector.

How does basic economic theory explain the decision?

Motivating example II

We write a model that includes:

- If static, utility function over consumption and leisure in each sector,
- Optimisation over the two options.
- If dynamic model, maybe budget constraint with non-labour income, that determines reservation wage;
- then, optimisation under constraint.

From micro-theory, we can learn about the preferences by looking at the choice. The preference for one sector or the other is said to be **revealed by the choice**.

Motivating example III

Traditionally, how do we empirically estimate the demand function, say for the public sector, in the population? For example, elasticity of the demand for public sector jobs to offered wage? Which type of data do we use?

Motivating example IV

Empirical analysis that includes:

- Use labour force survey (LFS) or administrative data,
- Use information about labour force status (sector of employment),
- Obtain socioeconomic characteristics of individuals (gender, age, education, test scores, etc.),
- We need to construct a predicted value of income in each sector.

Motivating example V

How can we construct (exogenous) measures of expected incomes? Which assumption(s) do we need to make?

Motivating example VI

With the Labour Force data:

- Look at similar individuals already on the market (gender, age, education, ability).
- Make the assumption that individuals have 'rational expectations' conditional on the information set.
- Make the assumption that we know what is important in the information set.

Motivating example VII

The main motivation of this line of research is that we are dissatisfied with these types of assumptions:

- Individuals do not necessarily have rational expectations about wages,
- Even if they had, we do not really know what they are condition on,
- Other beliefs may matter too: labour market tightness, arrival rate of job offers, job destruction rate, returns to search effort, etc.
- We cannot expect the individuals to have all these beliefs right, and they are not independent from each other. They are not all available in the type of data we have used traditionally.

The key problem is that we have constrained ourselves to a very limited set of data (LFS or administrative data), so-called *objective data*.

Motivating example VIII

As we will see, this causes important difficulties:

- We cannot disentangle the effect of beliefs and preferences based on the choices we observe. The same choice to work in a given sector is coherent with different sets of beliefs and preferences.
- For example, within the same gender-age-education-ability group:
  - some will not work for the public sector because they don't really like the public sector (**preferences**),
  - some will not work because they make a wrong prediction about the expected income (**beliefs**).

Motivating example IX

Imagine you could collect richer data on the individuals, what would you ask? What would be your concerns?

Motivating example X

Here are the two options we will discuss:

(1) Ask individuals directly what their beliefs are about the expected income.

(2) Ask individuals directly what they would do for different values of expected income.

Motivating example XI

While these approaches have been widespread in other social sciences, the economics profession has been late in adopting them (only in the 90s).

(1) There was (and still is sometimes) scepticism about the ability to measure beliefs accurately,

(2) There is good evidence that people do not do as they say.

The general objective of the lecture

In this lecture, I will:

1. review the discussion of Manski (2004), '*Measuring expectations*', which summarised the early days of an approach for asking and using individuals' subjective expectations.
2. show you an example Delavande (2008) that asks and uses **individuals' beliefs** about contraceptive method to understand the choice of method.
3. discuss the literature about **stated choices**, in particular stated choice experiments. I will use Meango and Girsberger (2025) as an example.
4. discuss an idea to combine revealed and stated preferences (time permitting).

It is not exhaustive! Can't be. It is just an introduction to the main concepts.

Manski (2004): Motivation and Background

Economists have long sought to **predict choice behaviour** through revealed preference analysis.

- Traditional approach: Infer decision rules from **observed choices** under various conditions (Samuelson, McFadden). For example, the quote from Varian (2014): *"Up until now, we were thinking about what preferences could tell us about people's behavior. But in real life, preferences are not directly observable:* ***we have to discover peoples' preferences from observing their behavior.****" (Chapter 7, my emphasis)*
- Problem: Observed behaviour can be consistent with many **combinations of preferences and expectations**.
- Result: Econometric models often rely on **strong assumptions**, especially regarding expectations (e.g., rational expectations).

Manski (2004): The Central Argument

Manski argues that predicting behaviour solely from choice data under **partial information** is problematic.

Advocates for directly **measuring expectations**—in the form of **subjective probabilities**.

- Allows relaxation or validation of assumptions.
- Provides richer empirical foundation.

Since the 1990s, surveys have increasingly **elicited probabilistic expectations** (e.g., job loss, income, voting).

Manski (2004)

Objectives of the Paper:

- Review the **limitations of revealed preference analysis**.
- Present the **case for eliciting subjective expectations** via surveys.
- Survey the emerging empirical literature and its **methodological foundations**.
- Highlight **key findings** and propose directions for **future research**.

Core Message

To improve behavioural prediction, economists should combine choice data with **measured expectations** rather than relying solely on assumptions.

An example: Schooling Choices and Perceptions I

Typical assumption: youth form expectations about the returns to education and choose schooling accordingly.

Empirical Problem: *Ad hoc* assumptions about how expectations are formed - may differ greatly.

- Freeman (1971): students expect to earn the mean income observed among earlier cohorts with the same major.
- Willis and Rosen (1979): youth condition on sex, armed forces status, and ability, and possess rational expectations.
- Manski and Wise (1983): expectations are based on a student's SAT score and the average SAT score at a college (without assuming rationality).

An example: Schooling Choices and Perceptions II

In a simple human capital model (Manski, 1993a), it is shown that the way youth use ability information to form expectations can crucially affect the interpretation of observed choices.

If youth do not condition expectations on ability, but the researcher assumes they do, the estimated model may misleadingly suggest that youth are indifferent to the returns to schooling.

- The analyst measures returns on ability that can be large,
- Yet, the individual does not consider ability in predicting returns: for example, they average over their peers.
- Thus, the analyst attributes the lack of response to ability as an indifference to returns, whereas it is a mistake in the forecast.

An example: Schooling Choices and Perceptions III

'Observing that persons forming expectations face the same inferential problems that empirical economists confront in their research, I find it generally implausible to assume rational expectations.' C. Manski

Another example: Identification in the Ultimatum Game I

The ultimatum game:
A *proposer* offers a division of a fixed sum K to a responder. The *responder* either accepts the offer (both players receive the proposed shares) or rejects it (both get nothing).

- $c \in [0, K]$ be the amount offered
- $d(c) = 1$ if the responder accepts and $d(c) = 0$ otherwise.
- The proposer's payoff is $y_1(c) = (K - c)d(c)$,
- and the responder's is $y_2(c) = cd(c)$.

Another example: Identification in the Ultimatum Game II

Consider six types of proposers, defined by two utility functions and three expectation rules.

Utility functions:

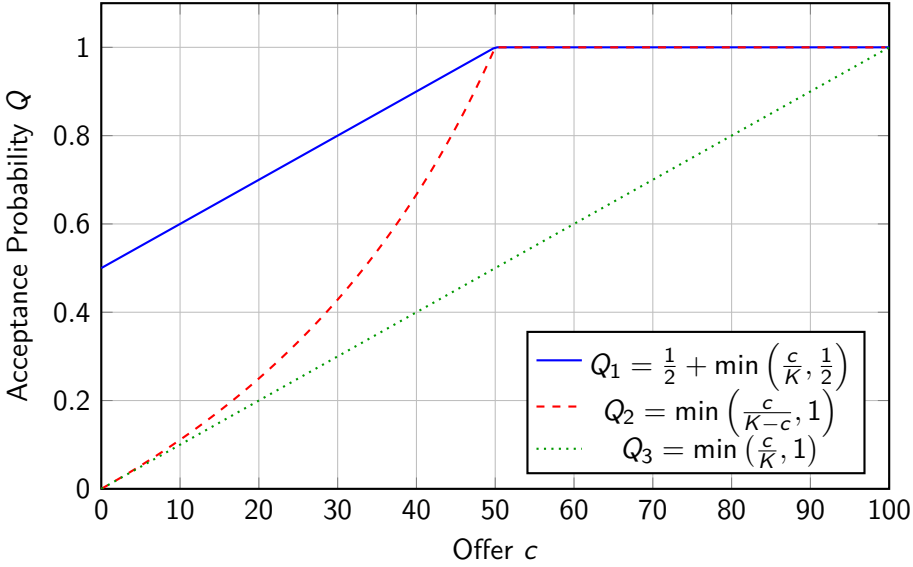
$$U_1(y_1, y_2) = y_1 = (K - c)d(c) \quad (1)$$
$$U_2(y_1, y_2) = y_1 \cdot y_2 = [(K - c)d(c)] \cdot [cd(c)] \quad (2)$$

The second utility function cares for the respondent.

Expectation rules for acceptance probability $Q_j[d(c) = 1]$:

$$Q_1 = \frac{1}{2} + \min\left(\frac{c}{K}, \frac{1}{2}\right), \quad Q_2 = \min\left(\frac{c}{K - c}, 1\right), \quad Q_3 = \min\left(\frac{c}{K}, 1\right)$$

Another example: Identification in the Ultimatum Game



Another example: Identification in the Ultimatum Game

Expected utility maximization yields optimal offers:

$$c(1,1) = \frac{K}{4}, \quad c(1,2) = \frac{K}{2}, \quad c(1,3) = \frac{K}{2}$$
$$c(2,1) = \frac{K}{2}, \quad c(2,2) = \frac{K}{2}, \quad c(2,3) = \frac{2K}{3}$$

Four of the six types choose $c = K/2$, an even split. Both fairness-oriented and self-interested agents may make the same offer if they share the same expectations.

Hence, observed choices in the ultimatum game do not uniquely identify preferences.

'I have concluded that econometric analysis of decision making with partial information ***cannot prosper on choice data alone****. However, combination of choice data with other data should mitigate the credibility problem and improve our ability to predict behavior.'* C. Manski

How to measure expectations: Attitudinal Research and Verbal Expectations

Attitudinal researchers have long used **verbal** questions to elicit expectations, such as: "Do you think the economy will improve?" or "How likely are you to lose your job?"

These responses often use vague phrases like "very likely" or "not too likely."

- **interpersonal incomparability** of verbal terms: what "fairly likely" means to one respondent may differ for another, for one question to another.
- **Coarseness** fails to capture rich uncertainty. For example, fertility question in the CPS:
  Looking ahead, do you expect to have any (more) children?
  Only allows "yes," "no," or "uncertain" responses

How to measure expectations: Cognitive Psychology and Probabilistic Expectations

Cognitive psychologists have extensively studied the elicitation of **subjective probabilities**, finding they offer significant advantages over verbal assessments.

- Numerical measures are more likely to be interpreted consistently.
- They allow researchers to test internal consistency using the laws of probability.

However, there's debate about whether people internally **represent beliefs numerically**: but the point is that people are **willing and able** to give numerical probabilities.

The Tversky–Kahneman work shows that people may rely on heuristics rather than Bayes' rule, but this challenges **rational expectations**, not the use of probabilities.

Probabilistic Expectations in Economics (Origins)

The idea of probabilistic expectations in economics dates back to **Juster (1966)**, who proposed eliciting purchase probabilities rather than yes/no intentions.

"Yes" means a subjective probability high enough to justify it - suggesting direct elicitation of probabilities is more informative.

Juster probability Question: 'Taking everything into account, what are the prospects that some member of your family will buy a [] sometime during the next [] months, between now and [] ?

Certainly, Practically Certain (99 in 100); Almost Sure (9 in 10); Very Probably (8 in 10); Probably (7 in 10); Good Possibility (6 in 10); Fairly Good Possibility (5 in 10); Fair Possibility (4 in 10); Some Possibility (3 in 10); Slight Possibility (2 in 10); Very Slight Possibility (1 in 10); No Chance, Almost No Chance (1 in 100).'

Juster found these to be better predictors of consumer behaviour.

Probabilistic Expectations in Economics

A more recent version of the scale: Survey of Economic Expectations - Introduction and Opening Question:

Now I will ask you some questions about future, uncertain outcomes. In each case, try to think about the whole range of possible outcomes and think about how likely they are to occur during the next 12 months. In some of the questions, I will ask you about the percent chance of something happening. The percent chance must be a number from 0 to 100. Numbers like 2 or 5 percent may be 'almost no chance,' 20 percent or so may mean 'not much chance,' a 45 or 55 percent chance may be a 'pretty even chance,' 80 percent or so may mean a 'very good chance,' and a 95 or 98 percent chance may be 'almost certain.' The percent chance can also be thought of as the number of chances out of 100.

Let's start with the weather where you live. What do you think is the percent chance (what are the chances out of 100) that it will rain tomorrow?

Probabilistic Expectations in Economics

Introducing this scale, or similar ones (based on a scale from 0 to 20, rather than 0 to 100) :

- allows to familiarise the respondents with the concept of probability of an event by giving a verbal meaning to the concept.
- helps making answer consistent and comparable across respondents and across questions,
- encourages the respondents to use the whole scale: A usual worry is that there is an excessive use of the values 0, 50, 100.

An example: Measuring Job Insecurity (Survey Questions) I

Manski and Straub (2000) designed survey questions to elicit subjective expectations about job security from employed respondents in the Survey of Economic Expectations (SEE).

Job Loss Question:

> *"I would like you to think about your employment prospects over the next 12 months. What do you think is the percent chance that you will lose your job during the next 12 months?"*

Search Outcome Question:

> *"If you were to lose your job during the next 12 months, what is the percent chance that the job you eventually find and accept would be at least as good as your current job, in terms of wages and benefits?"*

An example: Measuring Job Insecurity (Survey Questions) II

Quantile	Job Loss (%)	Search Outcome (%)
Mean	14.7	57.0
25th percentile	0	30.0
50th percentile	5	60.0
75th percentile	20.0	89.0

Table: Manski and Straub (2000), as reported in Manski (2004), Table I

The distribution of **Job Loss** is highly skewed—most respondents report low chances of losing their job.

In contrast, the **Search Outcome** distribution is more symmetric and dispersed, indicating varied beliefs about finding a comparable job.

The Handbook of Economic Expectations

The more recent development in measuring and using subjective expectations are presented in the handbook of economic expectations, here:

https://www.sciencedirect.com/book/9780128229279/handbook-of-economic-expectations

Most chapters (working paper versions) are available for free if you go on Google Scholar.

Predicting Choice Behaviour

Here are the two options we will discuss:

(1) **Augmenting the Revealed Preference** framework with data on expectations:
Ask individuals directly what their beliefs are, for example, about the state of the world or the returns to their choice.

(2) **Using a stated preference approach**:
Ask individuals directly what they would do in different situation.

Pill, Patch or Shot? (Delavande, 2008)

Main Topic: Understanding how women choose contraception methods based on their **subjective expectations**.

Which beliefs do you think matter? Which question would you ask?

Pill, Patch or Shot? (Delavande, 2008)

Main Topic: Understanding how women choose contraception methods based on their **subjective expectations**.

Which beliefs do you think matter? Which question would you ask?

Key Contributions:

- Uses probabilistic expectations to estimate a structural random utility model.
- Identifies preferences separately from beliefs.
- Applies model to assess policy interventions.

Why Study Subjective Expectations in Birth Control?

Unintended pregnancy is prevalent (about 50% of U.S. pregnancies). Choice of contraception is crucial, yet little is known about how women evaluate methods.

Research Questions:

1. What outcomes matter most in contraceptive choices?
2. How do subjective beliefs differ from population statistics?
3. How do expectations influence preferences and choices?
4. Can we simulate policy effects using these insights?

Theoretical Model – Setup

Women choose method m from choice set M_i to maximize:

$$\max_{m \in M_i} \int U_i(e_1, \ldots, e_n, c)\, dP_{im}(e_1, \ldots, e_n, c)$$

where e_j = binary outcomes (e.g., pregnancy, STD), c = cost, and P_{im} = subjective beliefs.

Assumptions:

- Utility additively separable across outcomes: we only need marginal probabilities.
- Linear in cost: we only need expected cost.
- Outcomes are binary: we only need to ask about the chance of the outcome occuring.

Random Utility Model

Let Δu_j be the marginal utility of avoiding outcome e_j.

Then the choice probability becomes:

$$\Pr(m) = \frac{\exp\left(\sum_j \Delta u_j P_{im}(e_j = 1) + aE_i(c_m)\right)}{\sum_{m' \in M_i} \exp\left(\sum_j \Delta u_j P_{im'}(e_j = 1) + aE_i(c_{m'})\right)}$$

This **mixed logit model** allows preferences Δu_j and price sensitivity a to be estimated using elicited expectations and observed choices.

Estimation by Maximum likelihood.

Data – Survey and Sample

Sample: 100 sexually active women aged 18–40 from Northwestern University and Truman College.

Survey Modules:

- Contraceptive choice and choice set
- Subjective probabilities for 8 outcomes
- Willingness-to-pay for 100% effective method

Respondents answered probabilistic questions with 0–100% estimates; 95% used rounding to nearest 5%.

Contraceptive Methods Considered

Respondents provided information about awareness, past use, and subjective beliefs for a wide range of contraceptive methods. Most women knew 13–14 methods. All were assumed to consider "no-method" as part of their choice set.

Methods included:

- Birth control pill
- Condom (male and female)
- Diaphragm, cervical cap, sponge, spermicides
- Depo-Provera (injectable)
- . . . ,
- "No-method" (includes traditional methods: rhythm, withdrawal)

Subjective Expectations – Outcomes Elicited

For each method in choice set, elicited:

- P(pregnancy)
- P(contracting STD)
- P(side effects, weight gain)
- P(partner/religion disapproval)
- P(interference with romance)
- P(return to fertility within 12 months)

Example question:

> *What do you think is the percent chance that you would get pregnant during the next 12 months if you were using [method]?*

This is a long survey! Six outcomes for eight different methods.

Table 4 – Subjective Beliefs: Pregnancy Risk on Pill

Subjective Belief (%)	All Respondents	Ever-users
0	9	7.6
1	18	24.5
2	15	11.3
3	11	13.2
5	18	18.9
10	5	5.7
20	3	1.9
40	2	1.9
50	2	1.9
60	2	1.9
70	1	1.9
100	1	1.9
N	**100**	**53**

Table: Delavande (2008), Table 4

Most respondents believe pregnancy risk on the pill is low (1–5%), though some report much higher values. Ever-users show similar beliefs to non-users.

Monetary Value of Outcomes – Table 7

Willingness to Pay to Reduce Outcome Probability by 5%

Outcome	WTP ($)
Getting pregnant	9.88
Contracting an STD	7.76
Partner disapproval	5.34
Interference with romance	1.24
Side effects	0.91
Religion disapproval	1.38
Weight change	-0.92

Avoiding pregnancy and STD risk have highest monetary valuation.

Marginal Effects of Changing Beliefs – Table 8

Predicted probability of choosing method after increasing one belief by 5%

Change in Belief (+5%)	Pill	Condom	Depo-Provera
Initial predicted probability	0.131	0.126	0.099
Pregnancy	0.082	0.079	0.061
Side effects	0.126	0.121	0.095
Weight change	0.137	0.132	0.103
Contracting an STD	0.091	0.087	0.068
Romance interference	0.124	0.119	0.093
Return to fertility	0.138	0.133	0.105
Partner disapproval	0.102	0.098	0.076
Religion disapproval	0.123	0.118	0.093
Cost +$5	0.104	0.100	0.078

Pregnancy and STD beliefs have strong marginal effects on method choice.

Preference Heterogeneity and Validation

Elicited Willingness-to-Pay:

- Median WTP for 100% effective method vs. 85% = \$30
- Implied from model: \$27.9 – strong coherence

Race-based heterogeneity:

- Whites: $\Delta u_{\text{pregnancy}} \approx -19.4$
- Non-Whites: $\Delta u_{\text{pregnancy}} \approx -7.8$

Model with stated WTP outperforms model with assumed failure rates.

Policy Simulations

1. Cost Subsidy: Making all methods free reduces predicted use of "no-method" by 22%.

2. Technological Change: Simulate improved STD protection for spermicides – predicted usage increases up to 150%.

3. Information Campaign: Setting Depo-Provera perceived pregnancy risk to 0.3% increases predicted use by up to 46%.

Method: Simulations use estimated model (Table 13) and updated beliefs.

Conclusion and Contribution

Delavande (2008) shows:

- Subjective expectations are measurable and meaningful.
- They are essential to identify preferences in discrete choice.
- Accounting for belief heterogeneity improves inference.
- Policy simulations based on these models can inform targeted interventions.

Main Critic: Beliefs are not exogenously set. They may depend on unobserved characteristics that also explain the choice of individuals.

The effects may not be causal, hence, counterfactual predictions may be wrong.

Two approaches to remedy this

First solution is to provide an exogenous shock to the beliefs:

- Randomly select some respondents and give them a piece of information on your variable of interest (see, for example Wiswall and Zafar, 2015).
- Problem: You cannot ensure that this does not also change beliefs about other factors.
- You can recover the reduced form effect of the information treatment on the outcomes.

Second solution, you can ask about choices in hypothetical situations.

- The researcher has the control over the characteristics presented to the respondents.
- Problem: intended choices, not actual choices.

The Stated Preference Approach I

Stated preference analyses aim to learn about human behaviour by asking people directly what they would do or choose in a hypothetical situation.

- **Choice-based conjoint** (leading for market research): present consumers with two or more products with different attributes, and ask them to pick one (see a historical review in McFadden, 2017).
- Same principle has been applied to several human capital investments or choices by social scientists (see discussions in Hainmueller et al., 2014; Maestas et al., 2023; Koşar and O'Dea, 2023; Giustinelli and Shapiro, 2023).

The Stated Preference Approach II

- **Strategic Survey Questions** present hypothetical scenarios that tend to be very elaborate, with the aim of retrieving precise model-based parameters (see an example in Ameriks et al., 2020b).
- Vignette analyses supplement choice alternatives with detailed, often visual story descriptions of alternatives (see an example in Adams-Prassl and Andrew, 2019).
- **Contingent valuation method** (environmental economics) valuing natural resources, where transactional markets typically do not exist and the non-use value of the choice alternatives is of interest (see, for example, Cummings, 1986).

The Stated Preference Approach III

The interest in the stated preference approach has accelerated recently, and it is increasingly used to:

(1) describe individual preferences by estimating the willingness-to-pay (WTP) for choice attributes McFadden and Train (2000); McFadden (2017)

(2) to conduct counterfactual policy analyses, that is, evaluating the impact of a policy that changes the distribution of choice attributes.

The Stated Preference Approach IV

The main advantages of the stated preference approach over the traditional revealed preference approach are that:

- the analyst can design experiments with rich variations in choice attributes,
- observe individuals' stated choices in counterfactual scenarios and explore preferences over policies never implemented before.
- It becomes easy to estimate causal effects of choice attributes because there is no concern of self-selection on unobserved characteristics.

The Stated Preference Approach V

Recent applications are wide-ranging and include education choices (Arcidiacono et al., 2020; Delavande and Zafar, 2019), mobility decisions (Gong et al., 2022; Koşar et al., 2022; Batista et al., 2025), long-term care and health investments (Kesternich et al., 2013; Ameriks et al., 2020a; Boyer et al., 2020), parental investments (Attanasio et al., 2019; Almås et al., 2024), marriage preference (Adams-Prassl and Andrew, 2019; Low, 2024), occupational choices (Maestas et al., 2023; Wiswall and Zafar, 2015, 2018), retirement decisions (Ameriks et al., 2020b; Giustinelli and Shapiro, 2024), voters' preference toward asylum seekers (Bansak et al., 2016), and irregular migration (Méango and Poinas, 2023).

Almås et al. (2024) predict that 'stated preference and answers to hypothetical questions' will become part of the normal toolkit in economics.

An example a stated Choice experiment I

Meango and Girsberger (2025): Examine the preference of elite school students in Côte d'Ivoire for the public or the private sector.

Try to understand whether there is competition for talent, and how costly it is for the private sector.

- Survey conducted in February 2024 at two elite schools,
- 587 valid interviews, one-third women.
- Contains beliefs data on the perception of the labour market.
- Contains a stated choice experiment to understand the preference for attributes of each sector.

Preference for the public sector I

If it was your study, which beliefs would you elicit?
How would you design the stated choice experiment?
What would be your concerns?

Preference for the public sector II

Eliciting Beliefs about the labour market attributes

Imagine 20 young Ivorians your age, i.e. [age variable]. Imagine they have an education similar to yours, i.e. [degree and field of study], all [male / female] like you, with similar family backgrounds. Think of these young people as they enter the job market. Out of these 20 young people, how many do you think will receive at least one job offer in the public sector?

Eliciting Beliefs about wages on the labour market

Imagine that these 20 young people each receive a job offer in the public sector. These job offers may vary according to the administration or public company offering them. We'll show you categories representing salary ranges, and ask you to allocate the twenty job offers to each range.

Preference for the public sector III

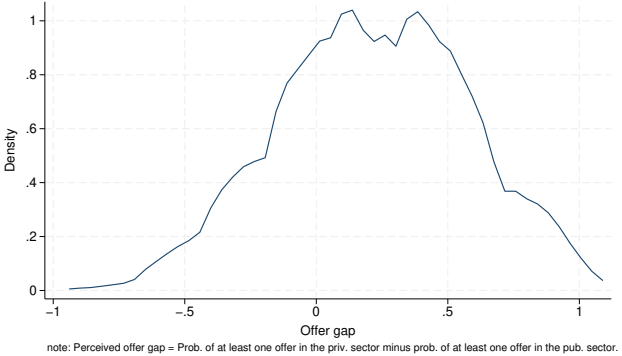
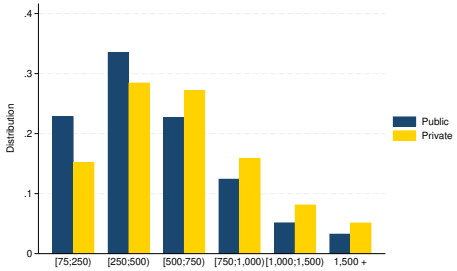


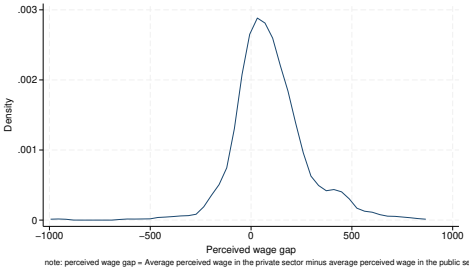
Figure: Perceived offer gap distribution

Note: The figure represents the probability distribution of the average perceived offer gap between two sectors. The offer gap is defined as the perceived probability of receiving a private sector offer minus the same perceived probability for a public sector offer.

Preference for the public sector IV



(a) Perceived wage distribution by sector



(b) Average perceived wage gap

Figure: Perceived wage by sector and average perceived wage gap

Note: In Panel (a), each bar represents the reported a probability of a wage offer following within a given bin by sector averaged on the sample. The intervals are in 1,000 CFA Francs. Panel (b) represents the probability distribution of the average perceived wage gap between two sectors.

Preference for the public sector V

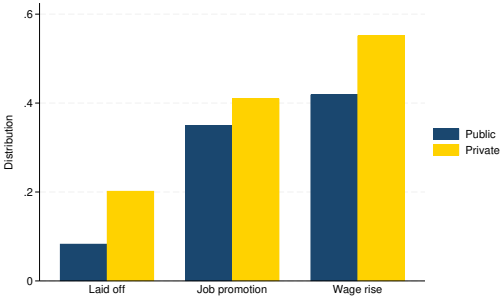
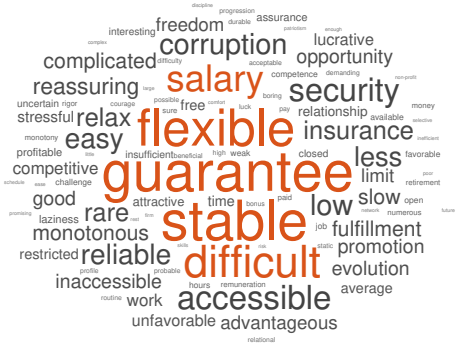


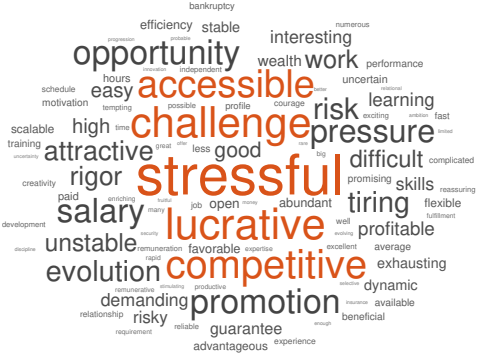
Figure: Other perceived characteristics by sector

Note: Each bar represents the reported probability averaged on the sample. 'Laid off' represents the probability of being laid off within two years, 'job promotion', the probability of being promoted to a more senior position, 'wage rise', the probability of receiving a 20 percent wage rise within two years.

Preference for the public sector VI



(a) Verbal description of the public sector



(b) Verbal description of the private sector

Figure: What students think about each sector in three words.

Note: The word cloud represent the words used by the students to describe each sector. Each respondent could use three words.

Preferences over sectors - Ranking

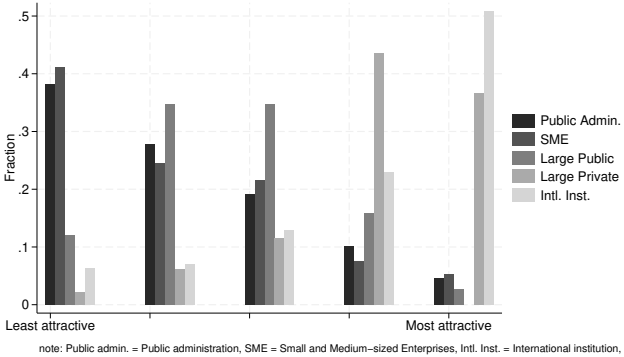


Figure: Attractiveness of sectors

Note: The figure represents the average ranking of employers. Ranks go from 1 (Most attractive) to 5 (least attractive). International institutions and large private firms are, in general, the most attractive, SMEs and public administration are the least attractive.

The Choice Experiment I

Choice experiment instructions.

Now we're going to show you a **series of hypothetical scenarios**. In these scenarios, you receive **two job offers** for a position that corresponds to your degree. One offer comes from the **public sector**, the other from the **private sector**.

In these scenarios, these two offers differ in the starting salary they offer, the number of hours of work expected, the risk of losing the job, and the progression in position.

The Choice Experiment II

For example, let's compare these two offers:

Offer 1:
Your employer: Public administration
Starting salary 500,000
Number of hours per week 40
Risk of losing job within two years 2/100
Number of people who become managers with salary +20%: 2/10

Offer 2:
Your employer: SME
Starting salary 800,000
Number of hours per week 60
Risk of losing job within two years 10/100
Number of people who become managers with salary +20%: 1/10

The Choice Experiment III

We are going to ask you to **consider several such scenarios**. Considering all the factors that might affect your decision to enter the job market (e.g. family background, your parents' wishes, macroeconomic conditions), we'd like to know which offer you'd be most likely to choose.

We'll ask you how likely you would be to choose the **[randomize= private/public]** sector. Between different scenarios, the only variations are indicated by the scenario. For example, the level of responsibility or the number of days off in the public sector job remain the same. **Only the stated characteristics change.**

You can consider that **these offers are your only options,** and that you would not receive another offer within two years if you turned down both offers.

Q1 - Q5. Consider the following two offers, what is the chance that you will choose the offer from the [randomize= private/public] sector?

The Choice Experiment IV

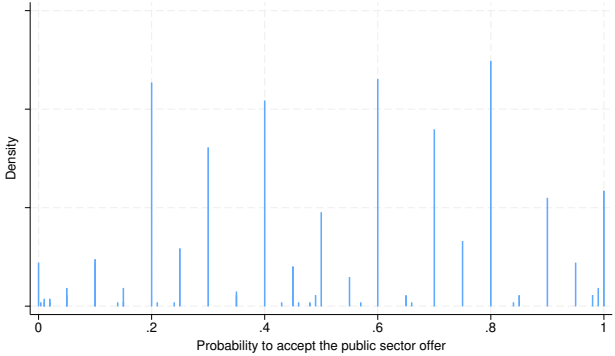


Figure: Probability of choosing the public sector for two identical offers

Note: The figure represents the reported probability of choosing a public sector offer, when offers are otherwise identical.

Random Utility Model with Stated Choices

A standard stated-choice model assumes each individual i chooses among J alternatives with utility:

$$U_{ij} = x_{ij}\beta_i + \varepsilon_{ij}$$

where x_{ij} are observed attributes (e.g., price, outage frequency) and ε_{ij} are unobserved i.i.d. Type I extreme value errors.

This leads to the conditional choice probability:

$$P(y = j|x_i, \beta_i) = \frac{e^{x_{ij}\beta_i}}{\sum_{h=1}^{J} e^{x_{ih}\beta_i}}$$

Aggregating across the population yields the mixed logit model:

$$P(y = j|x_i, \theta) = \int \frac{e^{x_{ij}\beta}}{\sum_{h=1}^{J} e^{x_{ih}\beta}} f(\beta|\theta)\, d\beta$$

Why Use Elicited Choice Probabilities?

Stated choice assumes the respondent knows their full utility function.

Problem: In practice, scenarios are incomplete — respondents lack information they would have in real decisions.

Solution: Elicit *choice probabilities* instead of fixed choices. These allow respondents to express uncertainty over their actions.

Interpretation:

$$q_{ij} = \Pr\left[U_{ij} > U_{ik}, \forall k \neq j\right]$$

When $q_{ij} \in (0, 1)$, the respondent indicates uncertainty about what they would choose in reality. This adds information compared to binary responses (0 or 1).

Linear Mixed Logit with Elicited Probabilities I

Assume that each respondent believes $\varepsilon_{ij} \sim$ i.i.d. extreme value, then:

$$q_{ij} = \frac{e^{x_{ij}\beta_i}}{\sum_{h=1}^{J} e^{x_{ih}\beta_i}}$$

Taking log-odds relative to base option $j = 1$:

$$\log\left(\frac{q_{ij}}{q_{i1}}\right) = (x_{ij} - x_{i1})\beta_i$$

Linear Mixed Logit with Elicited Probabilities II

Approach of Blass et al. (2010):

Assume $\beta_i = b + \eta_i$ and η_i has mean zero:

$$E\left[\log\left(\frac{q_{ij}}{q_{i1}}\right)\middle| x\right] = (x_{ij} - x_{i1})b$$

Estimate b by OLS or Median Regression (Least-Absolute-Deviation).

Approach of Wiswall and Zafar (2018):

Ask several scenarios to the same individual and estimate the model separately for each individual.
See, for example, Wiswall and Zafar (2018): 16 scenarios, Ugalde (2022): 30 scenarios, Koşar et al. (2022): 22 scenarios.

Computing Willingness to Pay (WTP) in the Linear Mixed Logit

WTP Definition: Amount of money a student is willing to pay to reduce x by one unit, holding utility constant.

From utility model:

$$U_{ij} = \gamma_i \mathrm{Wage}_j + \mathrm{Attributes}_j' \beta_i + \varepsilon_{ij}$$

where β_i is a vector of utility parameters associated with the attributes - recall Delavande (2008):

WTP for a one-unit reduction in a given attribute attr:

$$\mathrm{WTP}_{\mathrm{attr}} = -\frac{\beta_{i,\mathrm{attr}}}{\gamma_i}$$

Meango and Girsberger (2025): Just ask them twice ...

Meango and Girsberger show that if one only wants to understand the *population distribution of WTP* (not the individual values) for specific attributes, it is enough to present **two scenarios**.

The main point is to randomise the scenarios across individuals.

The paper also shows that there is no need for the linear utility or the i.i.d. extreme value assumption.

This allows for significantly **shorter surveys** and **more flexible decision models**: for example, how much I enjoy a high income can depend on having enough leisure time at my disposal.

MG2025: Conceptual framework I

Consider an economic agent i, a **binary** choice alternative 0 or 1, and **two consecutive periods**: a time of preference elicitation (today) and a time of decision (tomorrow).

The utility of agent i in option $d \in \{0, 1\}$ is given by:

$$U_d(y_d, z_d, \eta_i^*).$$

The utility depends on a *numeraire* that we will interpret here as expected income (y_0, y_1) in option 0 and option 1 respectively. We assume that the utility is continuous and strictly increasing w.r.t (y_0, y_1).

The vector $z = (z_0, z_1)$ represents further choice characteristics that can be manipulated within a hypothetical choice experiment.

We will use the shorthand $x := (y_0, y_1, z)$ and denote the support of x by $\mathcal{X}$.

MG2025: Conceptual framework II

The random vector η^* represents the unobserved individual heterogeneity at the **time of decision**. It represents possibly unobserved characteristics of individual i and unobserved choice attributes that matter in the decision, but are neither specified nor altered by the choice experiment.

They are never observed by the analyst, but are known by the agent at the time of decision (tomorrow). This could include a **private taste** for one of the two options or some private information about the returns to choosing one option over the other.

MG2025: Conceptual framework III

At the time of decision, i chooses between option 0 and option 1 based on a rule:

$$D_i(x) = 1\{U_1(y_1, z_1, \eta^*) \geq U_0(y_0, z_0, \eta^*)\} \tag{3}$$

$D_i(x)$ is the choice of individual i, when the first set of choice attributes are exogenously set to x.

MG2025: Conceptual framework IV

Noting that the utility is continuous and strictly increasing in income, this is equivalent to the threshold-crossing rule:

$$D_i(x) = 1\{S(x, \eta_i^*) \geq 0\}, \text{ where} \tag{4}$$

$$S(x, \eta_i^*) = y_1 - R(y_0, z, \eta_i^*), \text{ and} \tag{5}$$

$$R(x, \eta_i^*) = U_1^{-1}(U_0(y_0, z_0, \eta^*), z_1, \eta^*) \tag{6}$$

$R(x, \eta_i^*)$ is the reservation wage: The agent chooses option 1 if and only if the offer wage y_1 exceeds the reservation wage (given the choice attributes). It satisfies: $U_1(R(x, \eta_i^*), z_1, \eta_i^*) = U_0(y_0, z_0, \eta_i^*)$.

$S(x, \eta_i^*)$ is i's returns to choosing option 1 over option 0 at the time of decision, when the choice attributes are described by (x, η_i^*). It satisfies $S(y_0, R(y_0, z, \eta^*), z, \eta^*) = 0$.

MG2025: Conceptual framework V

Due to the time gap between the time of elicitation and the time of decision, and since η_i^* is not specified by the scenario, i does not know the realisation of η_i^*, although they might know some subcomponent.

The model assumes that i entertains a probabilistic distribution over their reservation wage and returns based on their available information, characterised by a vector η_i.

Define for any r, s in $\mathbb{R}$:

$$\begin{aligned} F_{R,i}(r;x) &:= \Pr(R(x,\eta_i^*) \leq r|\eta_i). \\ F_{S,i}(s;x) &:= \Pr(S(x,\eta_i^*) \leq s|\eta_i). \end{aligned} \tag{7}$$

The perceived distributions $F_{R,i}$ and $F_{S,i}$ will be the main object of analysis.

Conceptual framework: Elicitation

Assumption (Ceteris paribus)

Presented with scenario X, respondent i states:

$$P_i = \Pr(S(X, \eta_i^*) \geq 0 | \eta_i) = 1 - F_{S,i}(0) := m(X, \eta_i). \tag{8}$$

They do not infer new unspecified attributes as the specified attributes change (**?**).
We'll come back to it.

Assumption (Independence)

For any individual i in the population, $X_i \perp\!\!\!\perp \eta_i$ and has support $\mathcal{X}$.

Achieved **by design** in most stated preference experiments.

Conceptual framework: The key Idea

I observe $F_{S,i}(0; X) = m(X, \eta_i) = 1 - P_i$.

I need $F_{S,i}(s; x)$, for any s and x, to characterise the return and the resolvable uncertainty.

How do I get it?

Conceptual framework: The key Idea

I observe $F_{S,i}(0; X) = m(X, \eta_i) = 1 - P_i$.

I need $F_{S,i}(s; x)$, for any s and x, to characterise the return and the resolvable uncertainty.

How do I get it?

Lemma (Méango and Girsberger (2025))

Under Assumptions 'Independence' and 'Ceteris Paribus', the following holds:

$$F_{S,i}(s; x) \quad = \quad 1 - m(y_0, y_1 - s, z, \eta_i), \text{ for any } s \text{ such that } y_1 - s \in (0, y_{\max}). \quad (9)$$

Conceptual framework: The key Idea

It works because S is linear in y_1.

$$\begin{aligned}
F_{S,i}(s;x) &= \Pr(S(x,\eta_i^*) \leq s|\eta_i), \text{ by definition.} \\
&= \Pr(y_1 - R(x,\eta_i^*) \leq s|\eta_i), \text{ by definition of } R(.) \\
&= \Pr((y_1 - s) - R(x,\eta_i^*) \leq 0|\eta_i), \\
&= \Pr(S(y_0, y_1 - s, z, \eta_i^*) \leq 0|\eta_i) \\
&= 1 - m(y_0, y_1 - s, z, \eta_i), \text{ by definition of } m(.).
\end{aligned}$$

Other explanation:
The ex ante ante is just **a translation of the reservation wage** ($S_i = y_1 - R_i$).
Each time the respondent gives a probability of acceptance, they reveal information about their reservation wage.

Identification of the distribution of Quantiles

But we don't know $m(x, \eta)$!

η is possibly infinite-dimensional unobserved heterogeneity (Fixed-effect estimation as in Wiswall and Zafar (2018) doesn't work).

Never mind! We know its quantiles.

$$Q_{P|X}(\tau|X) = Q_{m(X,\eta)|X}(\tau|X) = Q_{m(X,\eta)}(\tau)$$

We can understand how quantiles of $S(.)$ are distributed in the population.

Identification of the distribution of Quantiles

We are interested in $F_{S,i}$ or equivalently in $F_{S,i}^{-1} := Q_{S,i}$. We will characterise:

$$\Pr\left[F_{S,i}(s;x) \geq \tau\right] = \Pr\left(Q_{S,i}(\tau;x) \leq s\right) = F_{Q,S}(s;\tau,x)$$

The proportion of the population such that their τ-quantile of S is lower than a given value s.

Identification of the distribution of Quantiles

We are interested in $F_{S,i}$ or equivalently in $F_{S,i}^{-1} := Q_{S,i}$. We will characterise:

$$\Pr\left[F_{S,i}(s;x) \geq \tau\right] = \Pr\left(Q_{S,i}(\tau;x) \leq s\right) = F_{Q,S}(s;\tau,x)$$

The proportion of the population such that their τ-quantile of S is lower than a given value s. We can show that:

$$F_{Q,S}(s;\tau,x) \quad = \quad \int_0^1 I\left\{Q_{P|X}(a|y_0, y_1 - s, z) \leq 1-\tau\right\} da$$

One can translate the conditional quantiles of P into quantiles of $S(.)$.

Visual preview

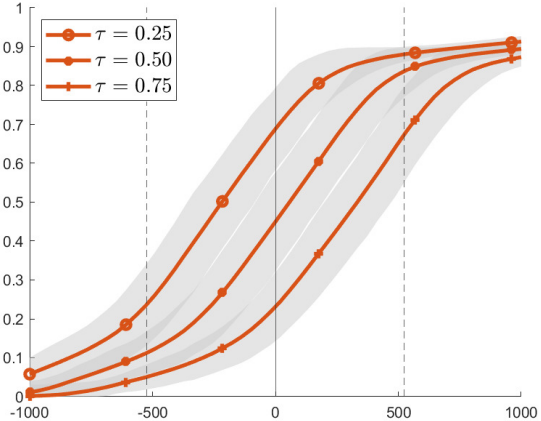


Figure: Distribution of quantiles $\widehat{F}_{Q,S}\left(.;\tau,x\right),\tau = 0.25, 0.50, 0.75$ for individual-specific perceptions and equating the perceived wage to the wage in the public sector.

WTP for specific attributes in the student sample I

How much would people pay for closing the gap between the public and the private sectors?

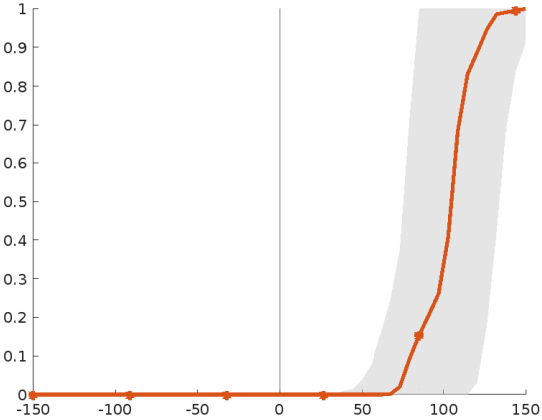
The starting point is identical job offers that are equal to the typical private sector offers.

Panel (a) changes the probability of being laid off in the public sector from the perceived average in the private sector (0.20) to the perceived average in the public sector (0.08).

Panel (b) changes the perception of the probability of obtaining a job promotion in the public sector from the average perceived level in the private sector (0.4) to the same average for the public sector (0.30).

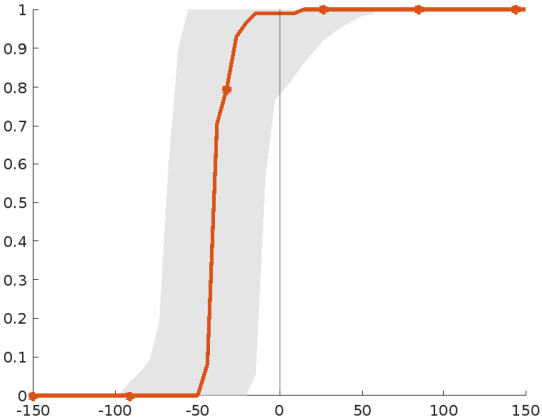
Panel (c) changes the working hours in the public sector from 45.8, as in the private sector, to 41.4, as in the public sector offer.

WTP for specific attributes in the student sample I



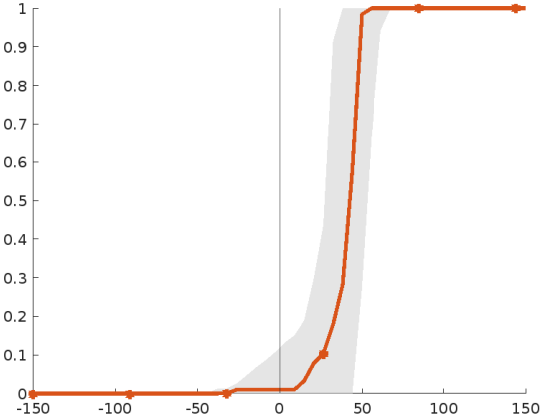
(a) Being laid off

WTP for specific attributes in the student sample II



(b) Obtaining a job promotion

WTP for specific attributes in the student sample III



(c) Working hours

Figure: Mean WTP, $mWTP$, for choice attributes : $\Pr\left(\mu_S(x + h\boldsymbol{e}_j, \eta) - \mu_S(x, \eta) \leq y\right)$

The Elephant in the Room

People do not do as they say.

This is known as the **hypothetical bias**.

- Encouraging evidence that carefully designed stated preference elicitation has predictive power for actual choices (e.g. Hurd, 2009; **?**; de Bresser and van Soest, 2019; Arcidiacono et al., 2020) and yield similar preference estimates as actual choice data (e.g. Mas and Pallais, 2017; Wiswall and Zafar, 2018)
- Concerns remain regarding systematic biases of stated preference data. See, for example, Murphy et al. (2005) and **?** and more recently Haghani et al. (2021a,b).

Our two approaches to learn about people behaviour

Revealed Preference approach (RP)

What people *choose* in real-life decisions.

(+) Relates to objective decisions,

(-) Only 'historical' decisions,

(-) Difficult to do causal inference, often because of unobserved heterogeneity.

Stated Preference approach (SP)

What people *say they would choose* when presented with hypothetical scenarios.

(-) Subjective decisions: hypothetical bias

(+) Rich, even never-experienced-before variations of choice attribute in

(+) Exogenous variations, i.e., easy causal inference.

Some motivation: Combine both approaches!

How to combine and perform causal inference is not so clear!

Attempt n: Van der Klaauw (2012); Wiswall and Zafar (2021):

- elicited behavior accurately portray optimal future behaviour conditional on current information. Any bias comes solely from biased information.
- problematic: large literature on present bias and hyperbolic discounting (see, e.g. Rubinstein, 2003).

Attempt $n+1$: Bernheim et al. (2021):

- The difference between stated and revealed demand function is stable across treatment and control groups.
- Problematic: strong restriction on treatment!

Some motivation

Henry, Meango, Mourifie (WIP): Stated preferences are useful:

- not because they match actual choices,
- but insofar as they provide valuable information on individual heterogeneity.

An example: The wage elasticity of labour force participation for job-seekers.

Imagine there are two groups in the population: (1) the prone to work, (2) and the not so prone ('only if wage is right!').

The unobserved heterogeneity is precluding causal inference.

A stated experiment varying wages = information that would otherwise not be observed.

- first group will report high propensity in **all** scenarios,
- second will report low propensity in general, but high **if wage is high**.

Econometric framework: Revealed preference

Two consecutive periods: a time of preference elicitation and a time of decision.
The economic agent i. The choice set $\mathcal{D}$ (discrete or continuous).

At the time of decision, i chooses one option in $\mathcal{D}$ based on a decision rule:

$$\begin{aligned} D_i(x) &= g(x, \eta_i, \nu_i) \\ D_i &= D_i(X_i) \end{aligned} \tag{10}$$

$D_i(x)$ represents i's choice, when the choice attributes are characterised by a vector x, D_i is the actual choice, as observed by the analyst at the time of decision.

Revealed preference

At the time of decision, i chooses one option in $\mathcal{D}$ based on a decision rule:

$$\begin{aligned} D_i(x) &= g(x, \eta_i, \nu_i) \\ D_i &= D_i(X_i) \end{aligned} \tag{11}$$

$\eta_i \in \mathcal{H}$, a vector space (low dimension), captures:

- unobserved traits or;
- i's private information set.

$\nu_i \in \mathcal{V}$, a vector space, captures:

- the *resolved uncertainty*, unknown to i at the time of elicitation, but is revealed at the time of decision (see a discussion in Blass et al., 2010)

In usual discrete choice models, the identifying restriction is $(\nu, \eta) \perp\!\!\!\perp X$.

(Probabilistic) Stated Preference

Prior to their decision, at the time of elicitation, i is asked to state their preference over different choice alternatives in hypothetical scenarios denoted by t.

In each scenario t, $X_{it} = x_{it}$ and i reports:

$$P(x_{it}) := P_{it} = \int_{\mathcal{V}} \tilde{g}\left(x_{it}, \eta_i, v\right) dF_{\tilde{\nu}|X_i,\eta_i}(v|x_{it}, \eta_i) \tag{12}$$

where:

- $F_{\tilde{\nu}|X_i,\eta_i}(v|x_{it}, \eta_i)$: perceived distribution of resolvable uncertainty at the time of elicitation;
- $\tilde{g}$: the stated counterpart of g in the stated experiment.

i can be wrong about the future ($F_{\tilde{\nu}} \neq F_{\nu}$) or their own behaviour ($\tilde{g} \neq g$).

Stated preference

Key assumptions about η_i: *stable* characteristics of individual i.

(1) stable across hypothetical scenarios, unaltered by the survey experiment: we have true *ceteris paribus* experiments.

(2) stable across SP and RP, i.e., affects both the stated and actual choice: Individuals consider, for example, their unobserved taste for work when answering the stated choice experiment.

Link with the *latent unconfoundedness* assumption **?**. Here, we have 'very similar outcomes' and allow for possibly multidimensional heterogeneity.

Objects of interest

Define the actual or objective demand function by

$$m(x, \eta) := \mathbb{E}(D(x)|\eta).$$

The counterpart in the stated choice experiment, the stated demand function, is

$$\tilde{m}(x, \eta) := \mathbb{E}(P(x)|\eta) = P(x)$$

The analyst's interest is in functionals of these objects, like the average structural function, ASF:

$$\mu(x) := \int_{\mathcal{H}} m(x, n) dF_\eta(n) \tag{13}$$

Identifying assumptions: conditional independence + support

One key identifying assumption for this counterfactual quantity is the following:

Assumption (Conditional independence)

Conditional on η_i, the vector of potential outcome is independent of X_i, that is,

$$(D_i(x))_{x \in \mathcal{X}} \perp\!\!\!\perp X_i | \eta_i \text{ or equivalently: } \nu_i \perp\!\!\!\perp X_i | \eta_i. \tag{14}$$

In addition, the support of η does not change with X.

Interpretation: Endogeneity is the result of η_i being associated with X and ν.
We use this assumption to match individuals on their stated preferences.

Identification result

Example of $d := \dim(\eta) = 1$. For any x, the function $\eta \mapsto \tilde{m}(x, \eta)$ is one-to-one mapping.

$$
\begin{aligned}
\mu(x) &=_{(1)} \int_{\mathcal{H}} m(x, n) dF_\eta(n) \\
&=_{(2)} \int_{\mathcal{H}} \mathbb{E}(D(x)|\eta) dF_\eta(n) \\
&=_{(3)} \int_{\mathcal{H}} \mathbb{E}(D_i|X_i = x, \eta_i = n) dF_\eta(n), \text{ by cond. ind. + support,} \\
&=_{(4)} \int_{[0,1]} \mathbb{E}(D_i|X_i = x, X_{i1} = \bar{x}, P_{i1} = p) dF_{X_{i1}, P_1}(\bar{x}, p)
\end{aligned}
$$

Identifying assumptions: one-to-one mapping

Assumption (Invertibility / One-to-one mapping)

Let $\dim(\eta) = d$.
(1) There are $T + 1 > d$ *scenarios, such that:*

$$P_{it} = \tilde{m}(X_{it}, \eta_i), t = 0, 1, \ldots, T \text{ with } \{X_{it} : t = 1, \ldots, T\} \perp\!\!\!\perp \eta, \tag{15}$$

In particular, we may have $X_{it} = \bar{x}_t$ *for all* i. *The support of* X_{i0} *is* $\mathcal{X}$. *The support of* $X_{it}, t \geq 1$ *is* $\mathcal{C}$, *is a compact included in the support of* X_i, $\mathcal{X}$.

(2) Furthermore, for any $(X_1, \ldots, X_d) \in \mathcal{C}^d$, *such that* $rank(X) = d$, *the mapping* $\eta \mapsto (\tilde{m}(X_1, \eta), \ldots, \tilde{m}(X_d, \eta)) \in [0, 1]^d$ *is globally homeomorphic, that is continuous and one-to-one.*

Identifying assumptions: one-to-one mapping

Example: multiplicatively separable log-odd model (Blass et al., 2010; Wiswall and Zafar, 2018; Koşar et al., 2022; Low, 2024). It corresponds to:

$$\boldsymbol{P} = \Lambda\left(v(\boldsymbol{X}) + K(\boldsymbol{X})\eta\right) \tag{16}$$

where $\Lambda(x) = \exp(x)/(1 + \exp(x))$.

We have a global homeomorphism if $K(\boldsymbol{X})$ is $d \times d$ matrix with full rank that spans $\mathcal{X}^d$.

In this case,

$$\eta = K(X)^{-1}\left(\Lambda^{-1}(\boldsymbol{P}) - v(\boldsymbol{X})\right)$$

.

Identification result

Proposition

Suppose Assumptions 'Conditional Independence' and 'Invertibility' hold, and $T = d$ then $\mu^r(.)$ is identified:

$$\mu^r(x) = \int_{\mathcal{C}^T \times [0,1]^T} \mathbb{E}(D_i | X_i = x, \boldsymbol{X}_i^T = \boldsymbol{x}^T, \boldsymbol{P}_i^T = \boldsymbol{p}^T) \, dF_{\boldsymbol{X}^T, \boldsymbol{P}^T}(\boldsymbol{x}^T, \boldsymbol{p}^T). \tag{17}$$

where $\boldsymbol{X}_i^T = (X_{i1}, \ldots, X_{iT})$ and $\boldsymbol{P}_i^T = (P_{i1}, \ldots, P_{iT})$.

The fundamental idea: to match individuals on their stated preferences.

Conclusion

In this lecture, we have:

1. reviewed the discussion of Manski (2004), '*Measuring expectations*', which summarised the early days of an approach for asking and using individuals' subjective expectations.
2. looked at Delavande (2008) that asks and uses **individuals' beliefs** about contraceptive method to understand the choice of method.
3. discussed the literature about **stated choices experiments**, using Meango and Girsberger (2025) as an example.
4. discussed an idea to combine revealed and stated preferences (time permitting).

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